

Post-Graduate Common Admission Test (PG CAT)

Official Detailed Syllabus: M.Sc Physics (Test Code 612)

Program: M.Sc in
Physics

Level: Post-Graduate Entrance
Examination

Eligibility: B.Sc Physics Degree with
Mathematics

Parameter	Examination Specification
Total Questions	150 Objective Multiple Choice Questions
Duration	2 Hours (120 Minutes)
Marking Scheme	+3 marks for correct response; -1 mark for incorrect response
Core Focus	Undergraduate (B.Sc) Physics core curriculum with auxiliary Mathematics

1. Mathematical Physics

- Vector calculus: gradient, divergence, curl, line, surface, and volume integrals; Stokes, Gauss, and Green theorems.
- Linear vector spaces, matrices, Cayley-Hamilton theorem, eigenvalues and eigenvectors.
- Complex analysis: Cauchy-Riemann conditions, Cauchy's theorem and residue calculus.
- Differential equations: first and second-order linear ODEs, Frobenius method, Legendre, Hermite, and Bessel special functions.
- Fourier series, Fourier transforms, Laplace transforms, and basic probability theory.

2. Classical Mechanics

- Newtonian mechanics of single and multi-particle systems, conservation laws, centre of mass.
- Constraints, generalized coordinates, D'Alembert's principle, Lagrange's equations and applications.
- Central force problem, Kepler's laws of planetary motion, scattering in a central field.
- Rigid body dynamics: inertia tensor, Euler angles, Euler's equations of motion.
- Hamiltonian formulation: canonical equations of motion, Poisson brackets, small oscillations, normal modes.
- Special theory of relativity: Lorentz transformations, length contraction, time dilation, relativistic energy-momentum.

3. Electromagnetic Theory & Electrodynamics

- Electrostatics: Laplace and Poisson equations, boundary value problems, method of images, multipole expansion.
- Magnetostatics: Biot-Savart law, Ampere's circuital law, vector potential, magnetic dipole.
- Electrodynamics: Maxwell's equations in vacuum and matter, gauge transformations (Coulomb and Lorenz gauges), Poynting theorem.
- Propagation of EM waves in linear dielectric and conducting media, reflection, refraction, Fresnel formulas.

4. Quantum Mechanics

- Inadequacy of classical physics, wave-particle duality, uncertainty principle, wave packet.
- Schrödinger wave equation (time-dependent and time-independent), probability interpretation, operators and observables.
- One-dimensional potentials: infinite square well, finite square well, potential barrier, delta function potential, harmonic oscillator.
- Schrödinger equation in 3D: hydrogen atom, orbital angular momentum, spherical harmonics.
- Spin angular momentum, Pauli spin matrices, addition of angular momenta. Time-independent perturbation theory.

5. Thermodynamics and Statistical Physics

- Laws of thermodynamics, thermodynamic potentials, Maxwell's relations, phase transitions, Clausius-Clapeyron equation.
- Microstates and macrostates, phase space, ensemble theory (microcanonical, canonical, and grand canonical ensembles).
- Maxwell-Boltzmann statistics, equipartition theorem, ideal gas.
- Quantum statistics: Fermi-Dirac distribution, degenerate Fermi gas, Bose-Einstein distribution, Planck black-body radiation formula, Bose-Einstein condensation.

6. Solid State & Condensed Matter Physics

- Crystal structures, reciprocal lattice, X-ray diffraction, Bragg's law, Laue conditions.
- Lattice vibrations, phonons, specific heat (Einstein and Debye models).
- Free electron theory of metals, Drude-Lorentz model, electrical and thermal conductivity, Wiedemann-Franz law.
- Band theory of solids, Bloch's theorem, Kronig-Penney model, semiconductors, effective mass. Superconductivity: Meissner effect, Type I/II, BCS overview.

7. Electronics and Atomic/Nuclear Physics

- Semiconductor physics, p-n junctions, bipolar junction transistors (BJT), field effect transistors (FET), operational amplifiers (Op-Amps) and configurations.
- Digital electronics: Boolean algebra, logic gates, flip-flops, counters, registers.
- Atomic spectra: Bohr-Sommerfeld model, LS and jj coupling, Zeeman effect, Stark effect.
- Nuclear properties: binding energy, nuclear models (liquid drop, shell model), radioactive decay laws, nuclear reactions, elementary particles.